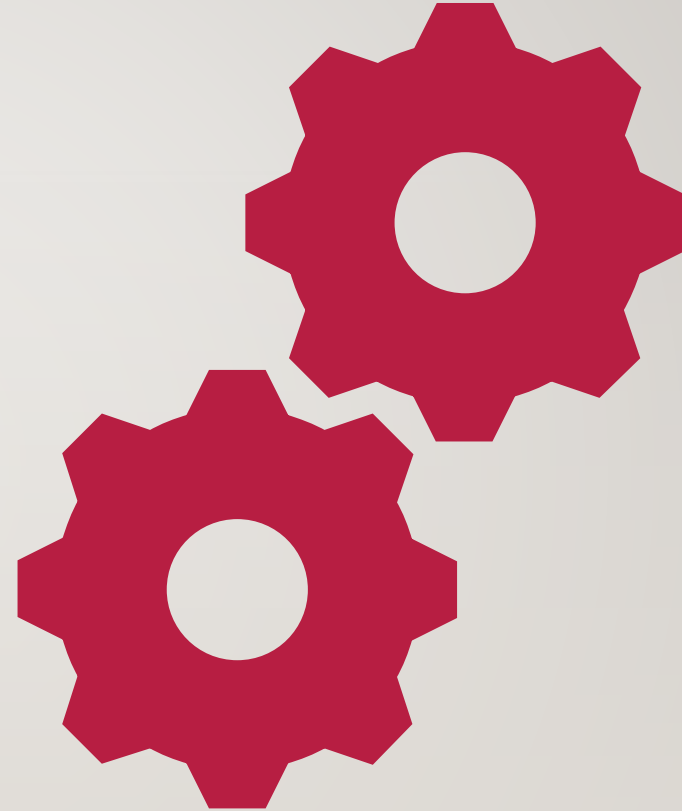


AUTOMATED RECALL MANAGEMENT SYSTEM FOR CAREFUSION/BD HEALTHCARE COMPANY

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WHAT IS ARMS

- ARMS is the Automated Recall Management System.
- We created (ARMS) because our Alaris and BD products were being recalled because of compatibility issues, reading errors.

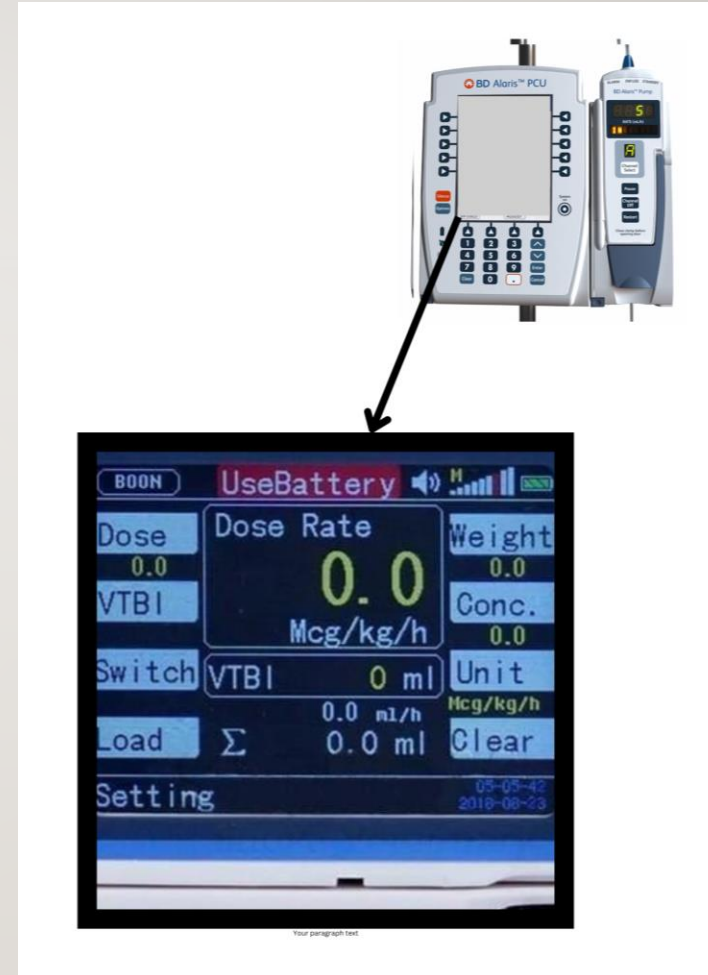
HOW DOES ARMS WORK

- ARMS will send out alerts to notify companies and customers which products are being recalled.
- New designs will be regulated with this system
- A list of compatible third-party products will be made
- Licensing and agreements for products will be managed through this systems

EXTERNAL QUALITY AGREEMENT SYSTEM

- Supplier quality team to assess external product design changes on a quarterly basis at a minimum.
- (eQAS) will also be integrated into the existing design control enterprise software such that both internal and external design changes are accessed via the same ways of working
- A new module will be added to design control to track and monitor designs and agreements.
- Also, flag which compatibility requirements are at risk given the status of the external agreements

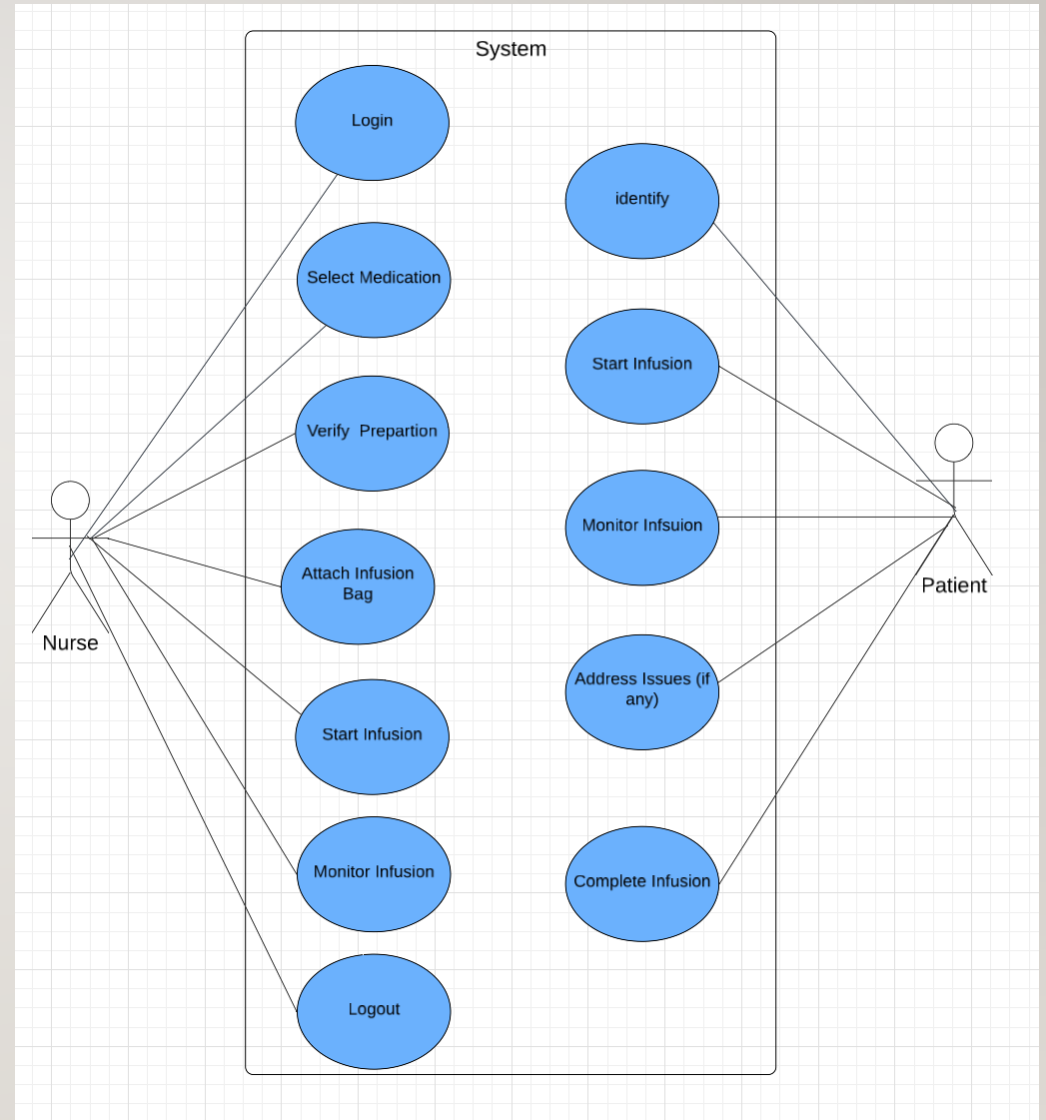
USER INTERFACE PROTOTYPE



USER INTERFACE

- The user interface is created for the Automated Recall Management System (ARMS) will focus more on simplicity, clarity, and basic functionality to ensure effective tracking and communication during the recall process.
- The interfaces first main component header section that will contain the company logo along with a navigation menu. This will also include a dashboard that will display notifications, responses received, responses pending action, and the total number of pumps replaced.
- The notifications section will display the logs of certain customers information like their name, date and time of notification sent, as well as their contact information. This section will also include options to resend notifications to customers who have not responded to the initial attempt at contact.
- The responses section in the UI will mainly display response status (acknowledged, pending) and the form completion status to have the pump replaced. This section will also include response tracking which will have color coding as a visual representation of response status.
- A settings area will also allow user management settings to be changed to the liking of an admin or operator.
- The last two sections included will be for prototype testing as well as QA monitoring. The prototype section will display a progression bar showing the different development stages of the project as well as detailed prototype testing reports and results. Lastly, the QA section will be available to include statistics like response rates, client list completeness, and the production status of new pumps.

USE CASES



RECALL NOTIFICATIONS

- **Test Case:**
- Verify that the Automated Recall Management System (ARMS) sends recall notifications and identifies impacted pumps to update the compatibility list.
- **Expected Results:**
- Every customer receives a recall notification.
- Identify affected pumps and update the compatibility list.

EMAIL NOTIFICATIONS

- **Test Case:**
- Ensure the ARMS sends recall notifications via email to at least 50 customers per minute.
- **Expected Results:**
- At least 50 customers receive a recall email from ARMS within a minute.

TEXT MESSAGE NOTIFICATIONS

- **Test Case:**
- Verify that ARMS sends recall notifications via text message to customers who opted for text notifications.
- **Expected Results:**
- Every opted-in customer receives a recall text from the ARMS system.

NOTIFICATION TRACKING SYSTEM

- **Test Case:**
- Ensure the tracking system identifies customers who have or have not filled out a response form and alerts the recall team if necessary.
- **Expected Results:**
- The system notifies the recall team of customers who did not fill out a response form.

EXTERNAL QUALITY AGREEMENT SYSTEM (DESIGN COMPATIBILITY)

- **Test Case:**
- Assess external product designs and notify the system admin and supplier quality team about their compatibility.
- **Expected Results:**
- The system alerts the teams when the design is compatible.

EXTERNAL BRAND COMPATIBILITY

- **Test Case:**
- Determine if an external brand's design is compatible or non-compatible, triggering an alert in the External Quality Agreement System (eQAS) and sending a notification to the team.
- **Expected Results:**
- Non-compatible designs are stopped from production and the team is alerted.

TIMELINE

- Updating Existing Database for New Pumps (1 week)
- Development will work as hard as possible to allow the current system room for the new pumps to be sent out. This should not be a hard task as the system has scalability and was set up for future sales.
- Recall Stage (1 Week to 2 Weeks)
 - In the first stages, the recall team will push the recall notifications to all hospitals and required parties. The recall team will have a small number of employees checking the recalls sent out against a list of all clients to ensure the recall has reached everyone. The coders will also be making sure the ARMS is emailing at least 50 customers a minute to make sure the entire client list is completed as fast as possible. For customers that have opted into the text option, a text will be sent directly to them including a link they can click to read more about the recall. Another small number of employees will be set up to receive the responses and log them into the system when clients fill out the response form. Hopefully all clients can respond within a week and by the end of two weeks all affected pumps will be returned.
- Brainstorming/Research Phase (1 to 2 Weeks)
 - Simultaneously, the entire Research and Development department and several outside sources will begin to break the pumps down and try to get an idea of what is creating the problems with the pumps. This will take 1 to 3 days hopefully. After the issue has been found, the rest of the time will be spent figuring out a full solution to the problem.
- Prototyping (1 week)
 - The engineers and R&D will begin taking the notes from the last phase and building a single prototype that fixes the issues presented. This prototype will then be added to the equipment to make sure the pumps are working correctly. If all is working correctly, we will begin to move to the production phase.

TIMELINE (CONT.)

- Production (3 weeks)
- The production team will begin making all the pump replacements. The production team will work overtime to ensure the pumps are completed as soon as possible.
- Testing (1 to 3 Days)
- After production has been completed, random pumps will be pulled to ensure a random sample batch and then be tested at several different hospitals. If all works properly, the full roll out will begin.
- Full Roll Out (2 Weeks)
- Replacements for all units will be sent out to clients. A notification will be sent out through the ARMS that instructs clients to notify us when they receive their new units. It will also give them instructions on how to dispose of the defective systems.
- Monitoring Quality Assurance (Ongoing)
- Going forward all Quality Assurance will be measured through the existing system. Teams will monitor the client list to make sure all clients have received up dated parts. The team will use the ARMS to continue to alert clients who have not responded to the recall or any other steps of the process. The old pumps production will be halted.

CONCLUSION

- “ARMS” saves our company \$1 million+ dollars in revenue because we will produce our own designs.
- External companies will now have to have agreements/licensing to pair with our products
- Old products are being recalled and consumers will be notified.